PROJECT THREE MILESTONE THREE: PROTOTYPING, DECISION MAKING AND DESIGN REVIEW #2

MILESTONE 3.1 – REFINED CONCEPT: INITIAL PROTOTYPE

Team ID: Mon-23

Complete this worksheet individually before coming to Design Studio/Lab A for Week 8.

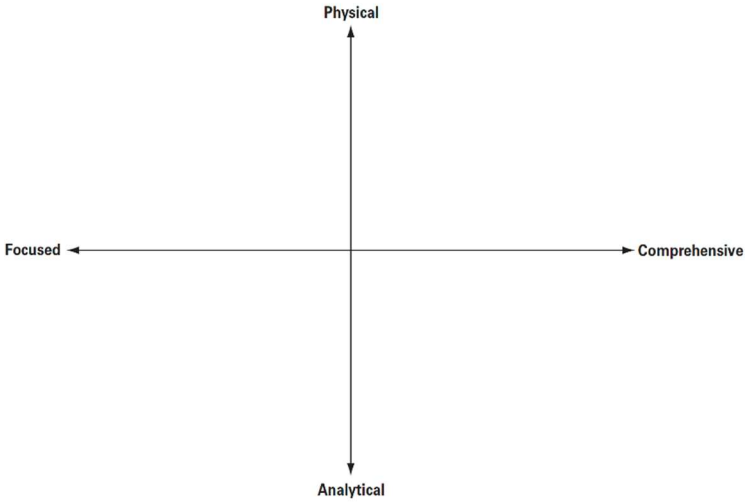
1. Write a small description of your initial prototype. Be sure to include what problem it aims to solve, how your initial prototype will be fabricated, and what functionality will be included and omitted in this initial prototype.
2. Classify whether your prototype is Physical or Analytical, and Focused or Comprehensive. Include the purpose of this prototype in the context of project 4 and the level of fidelity (low, medium, or high fidelity)
   - → **Physical vs. Analytical:** Physical prototypes are tangible artifacts that are created to approximate the final product. Analytical prototypes are non-tangible and represent the product using usually visual or mathematical models.
   - → **Focused vs. Comprehensive:** Focused prototypes implement only one or a few of the attributes of the final product. Comprehensive prototypes aim to implement most, if not all of the attributes of the final product.
3. Create a list of objectives and metrics for your initial prototype. There is no required amount of objectives or metrics, so long as the list is comprehensive.
4. Create a rough experimental plan for how you might test your prototype. Consider the methods you might use to test various objectives, how you will measure how effective each test proves to be and how realistic it would be to implement. This does not need to be detailed plan but should consider several of your objectives for the prototype.
5. Take picture(s) of your refined concept (initial prototype)
   - → Insert your photo(s) as a Picture (Insert > Picture > This Device)
   - → **Do not include more than *two* refined concept pictures per page**
   - → Include details on how concept was refined (what feedback was incorporated, what features are different than previous concept exploration, etc.)
   - → You can continue this process within the allocated time of the LabA/DS and seek feedback and discussions from your team members and/or the instructional team (IAIs, TAs, etc.).

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Write a short description of your initial prototype below.

The sidewalk indent detector utilizes the detection of the joints in sidewalks to determine if the user is currently walking on a sidewalk. This prototype is built using an Arduino microcontroller, a breadboard, 3 LEDs and a photoresistor, which acts as a stand-in for an IR/colour sensor. A white LED will be used to generate some light for the IR/colour sensor to recognize areas with high or low contrast, with low contrast areas representing the indents in the sidewalk. When an indent is detected, an orange LED blinks to show that the indent was detected. If an indent has not been found for quite some time (4 seconds), a blue LED turns on, signaling that the user is no longer moving on a sidewalk and may be in danger.

Indicate where your prototype falls on the scale below.



Kind of Prototype:

☐ Physical or ☐ Analytical

☐ Focused or ☐ Comprehensive

Purpose of Prototype: To determine whether the user is walking in a safe or dangerous area.

Level of Fidelity: Low to Medium

Include a list of objectives and metrics for your prototype below.

Objectives

- Provides effective feedback to the user when they are not walking on a sidewalk
- Accurately detects and identifies indents, even in low-light conditions
- Determines if the user is walking on a sidewalk or not

Metrics

- Time it takes the user to notice the alarm (seconds)
- Number of Missed Indents
- Number of False Alarms

Include a rough experimental plan on how you might test your prototype below.

Objective: Provides effective feedback to the user when they are not walking on a sidewalk

Metric: Time it takes the user to notice the alarm (seconds)

Test: Conduct 3 sets of 5-7 trials during the day, night with streetlights and night without streetlights. Each set would vary the LED's light intensity and the reaction time of the user to notice

the alarm will be recorded. The light intensity with the lowest average reaction time will be the best one.

Objective: Accurately detects and identifies indents, even in low-light conditions

Metric: Number of Missed Indents

Test:

Conduct 3-4 sets of 4 trials with each set testing a different sight LED (the white LED that is used to create high-contrast conditions for the IR/colour sensor) angle. Each trial would walk over a predetermined path with known indent locations. The missed indents per trial would be recorded and each angle's missed indent average will be determined. The most optimal sight LED angle will be based on the lowest average of missed indents.

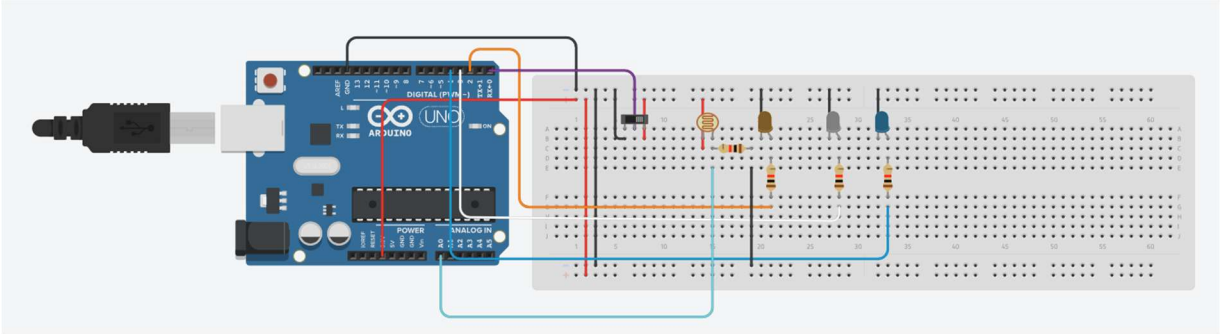
Objective: Determines if the user is walking on a sidewalk or not

Metric: Number of False Alarms

Test: Conduct 3 sets of 5 trials, each with a different time threshold. Each trial will be conducted on and off the sidewalk at a consistent speed. The number of false alarms would be measured per trial and an average would be calculated for each different time threshold. The time threshold with the lowest false alarm rate would be chosen.

Insert picture(s) of your refined concept (initial prototype) below.

Interfacing:



Code:

//NAME: Prabhjot Bhandal

//DATE: March 6th, 2025

//TITLE: 1P13 P3 MS3: Indent Investigator

```
//VARIABLES
//BUTTONS + LEDS
int START_SWITCH = 0; //PURPLE
int INDENT_DETECTED_LED = 2; //ORANGE
int CONTRAST_LED = 3; //WHITE
int NOT_ON_SIDEWALK_LED = 4; //BLUE

//SENSORS
//PHOTORESISTOR, stand in for a colour sensor (contrast),
//assigned to pin A0, analog!!!
// <450 = Dark, Indent Detected
// >450 = Light, Sidewalk Detected

//LOGIC VARIABLES
//unsigned long is a data type that keeps its entire value
//and can only be positive.
unsigned long lastIndentTime = 0;



void setup()
{
  //START BUTTON
  pinMode(START_SWITCH, INPUT);
  //CONTRAST
  //NOTE: For interfacing Photoresistor, must connect another
  //resistor to it to ground it.
  pinMode(A0, INPUT);

  //LEDS
```

```
  pinMode(INDENT_DETECTED_LED, OUTPUT);
  pinMode(CONTRAST_LED, OUTPUT);
  pinMode(NOT_ON_SIDEWALK_LED, OUTPUT);

  //Starts up serial monitor for comments
  Serial.begin(9600);
}

void loop()
{
  //Starts internal timer, how long it has been running
  unsigned long currentTime = millis();
  Serial.println(analogRead(A0));

  //Once switch activated, circuit turns on
  if (digitalRead(START_SWITCH) == HIGH) {
    //Turn on sight LED
    digitalWrite(CONTRAST_LED, HIGH);

    //When indent detected
    if (analogRead(A0) < 325) {
      //Indent detected
      Serial.println("Indent Detected; Still on Sidewalk");

      //Reset Timer
      lastIndentTime = currentTime;
      digitalWrite(INDENT_DETECTED_LED, HIGH);
      digitalWrite(NOT_ON_SIDEWALK_LED, LOW); // OFF sidewalk LED
    }
```

```
    //When indent isn't detected
    else {
      //Turn off indent found led
      digitalWrite(INDENT_DETECTED_LED, LOW);

      //If the indent hasn't been found for awhile, greater than
      //4 seconds
      if ((currentTime - lastIndentTime) > 4000) {
        //Turn on not on sidewalk LED
        digitalWrite(NOT_ON_SIDEWALK_LED, HIGH);
        Serial.println("NOT ON SIDEWALK!");
      }
      else {
        digitalWrite(INDENT_DETECTED_LED, LOW);
        Serial.println("Away from indent.");
      }
    }

  }
  else {
    Serial.println("OFF");
  }

  //digitalWrite(LED_BUILTIN, HIGH);
  delay(1000); // Wait for 1000 millisecond(s)
  //digitalWrite(LED_BUILTIN, LOW);
  delay(1000); // Wait for 1000 millisecond(s)
}
```